NOC PERFORMANCE ANALYTICS

Ticket Management, Operational Performance and SLA Compliance | 2025–2026

DATA ANALYTICS CASE STUDY

Prepared by Jesús Toledo

This case study documents an analytics workflow from Databricks SQL source exploration through Power Query, data modeling, DAX measures, interactive reporting, and operational recommendations.

10,020	22.25 h	7,267	17.64%
Total tickets	Average resolution time	Closed tickets	Reported SLA compliance

Reporting note: the dataset covers 2025 through August 14, 2026. August 2026 is a partial month and should not be compared directly with complete months.

1. Executive Summary

NOC Performance Analytics analyzes 10,020 support tickets to understand ticket volume, operational workload, resolution time, and SLA outcomes. The report is organized into three Power BI pages: Executive Overview, Operational Performance, and SLA & Resolution.

Metric	Observed result	Interpretation
Total tickets	10,020	Overall dataset volume.
Average resolution time	22.25 hours	Average of available resolution-time values; missing values are not zero.
Closed tickets	7,267 (72.52%)	Most records are marked Closed.
Open tickets	971 (9.69%)	Requires backlog monitoring.
In-progress tickets	1,782 (17.78%)	Requires ownership and progress tracking.
Reported SLA compliance	17.64%	1,282 compliant out of 7,267 tickets classified as compliant or breached in the current model.

Executive takeaway

The strongest signals are low reported SLA compliance, long resolution times, and 2,753 open or in-progress tickets. These are descriptive findings: they indicate where to investigate but do not, by themselves, prove root causes.

2. Business Problem and Data Sources

Business problem

A network operations center needs a consistent view of ticket demand, resolution performance, and SLA outcomes. The central question is: **Where is workload concentrated, how quickly are tickets resolved, and where should the NOC investigate potential service-performance issues?**

Data source and tables

The source data was queried in Databricks SQL using the workspace's **default** schema. The primary reporting table is **tickets**, related to **equipos** through equipment_id. The workspace also contains **clientes** and **tecnicos**; these were not required for the current two-table model shown in the supplied screenshot.

The tickets table includes identifiers, creation and closure timestamps, status, priority, incident category, region/city, and resolution-time fields. The equipment table includes equipment ID, name, type, region, model, and installation date.

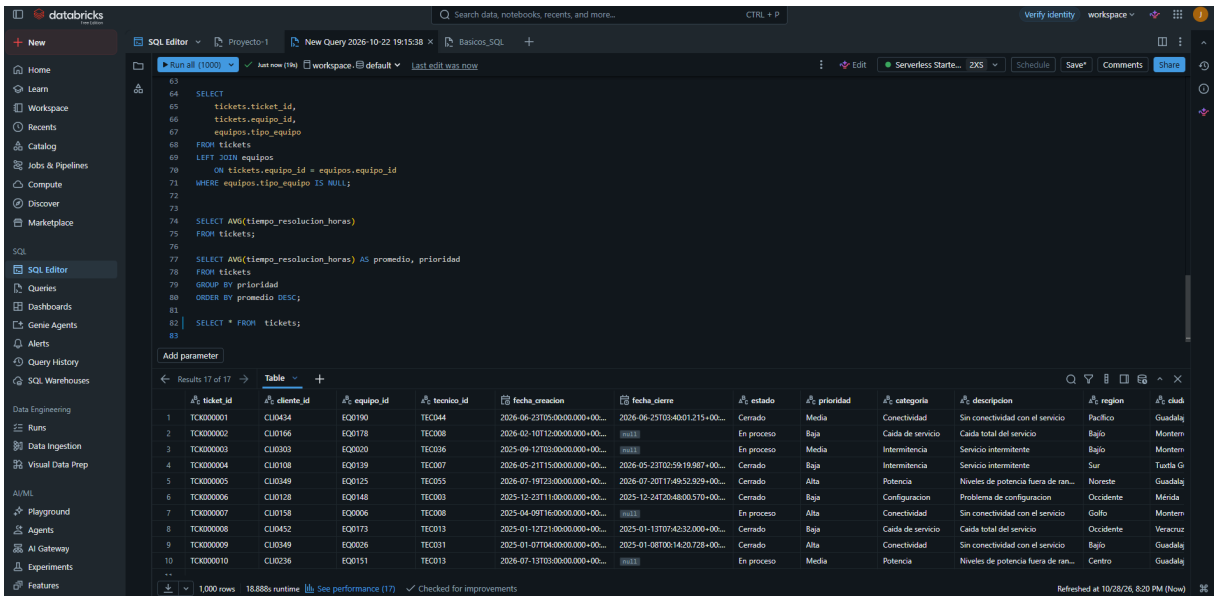


Figure 1. Databricks SQL Editor showing a query against the tickets table and sample ticket records. Connection credentials are not included.

Data quality notes

The source contains null closure dates and null resolution-time values. These should not automatically be treated as zero or as failed SLA cases. The report includes a 'No evaluated' resolution-time category. August 2026 is incomplete, ending on August 14.

3. Tools and Methodology

Tool	Role in the project
Databricks SQL	Source exploration and SQL queries to inspect ticket records, calculate summaries, and explore ticket/equipment relationships.
Power Query	Inspect imported columns, data types, null values, and table structure before modeling.
Power BI data model	Relate tickets to equipment using equipment_id, enabling analysis by equipment attributes.
DAX	Create reusable measures for total tickets, ticket statuses, average resolution time, and SLA calculations.
Power BI visuals	Present KPIs, monthly trends, status, priority, region, equipment type, incident category, resolution-time intervals, and SLA status.

Workflow

1. Source	2. Prepare	3. Model	4. Measure	5. Communicate
Databricks SQL	Power Query	Tickets + equipos	DAX measures	Power BI report

Data model

The model uses tickets as the fact table and equipos as a dimension, joined through equipo_id in a many-to-one relationship. A separate Medidas table organizes DAX measures.

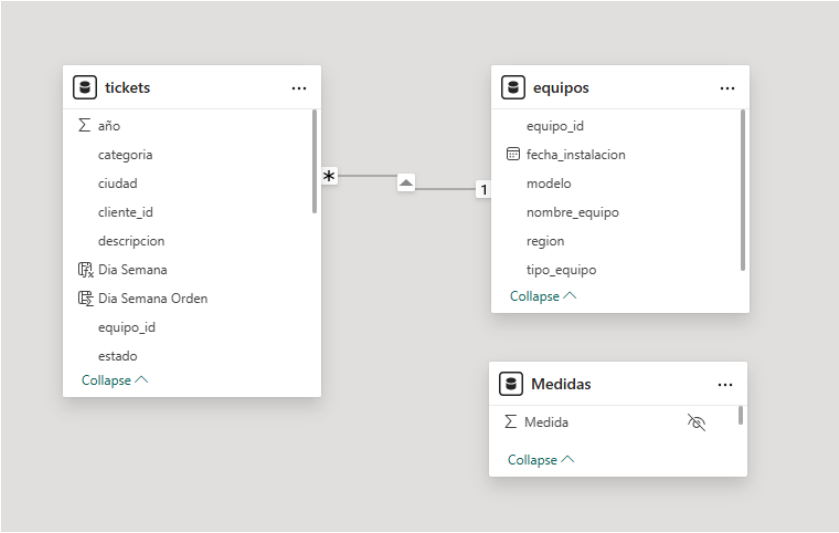


Figure 2. Power BI model showing the tickets-to-equipos relationship and a dedicated measures table.

4. Dashboard Structure

Page 1 — Executive Overview

Provides a high-level view of ticket volume, average resolution time, ticket status, monthly trends, and reported SLA compliance. It is designed for managers and operational leads.

Page 2 — Operational Performance

Explores workload across incident categories, equipment types, regions, priorities, and days of the week. The monthly priority matrix helps compare how demand changes over time.

Page 3 — SLA & Resolution

Examines SLA status, the distribution of resolution times, ticket status by priority, and monthly ticket counts alongside records with closure dates.

Design choices

The report uses a consistent teal theme (primary accent #4B9294; page background #F5FAF9), English titles and slicers, KPI cards, and chart types selected to answer operational questions.

Evidence note

The supplied Databricks and model screenshots are included as evidence of the source and model. The three dashboard screenshots were reviewed during project development, but this PDF version describes the dashboard pages in text rather than reproducing them as full-page images.

5. Key Findings

1. Ticket status and workload

Of 10,020 tickets, 7,267 are closed, 971 are open, and 1,782 are in progress. Open and in-progress tickets total 2,753, making backlog monitoring relevant.

2. Resolution time

Average resolution time is 22.25 hours. The interval chart shows 2,655 tickets in the 12–24 hour interval and 2,663 above 24 hours. Another 2,753 records are marked 'No evaluated' because a resolution time is not available for evaluation.

3. SLA compliance

The current classification contains 1,282 compliant and 5,985 breached tickets, producing 17.64% compliance among 7,267 classified tickets. Interpret this alongside the configured SLA rules and evaluation scope.

4. Incident categories

Connectivity has 2,483 tickets and Intermittency has 2,004. Together they account for 4,487 tickets (about 44.8% of the dataset), making them candidates for root-cause analysis.

5. Day-of-week volume

Friday has the highest displayed ticket volume (1,507), while Tuesday has the lowest (1,392). This suggests a pattern to investigate, not proof of a staffing issue.

6. Regional resolution time

Average resolution time by region ranges approximately from 21.72 to 22.81 hours in the displayed chart. Further segmentation may be needed to explain differences.

Interpretation guardrail

These are descriptive findings. High category volume does not prove a specific equipment failure is the root cause, and regional MTTR differences do not by themselves establish a performance problem.

6. Recommendations and Limitations

Finding / risk	Recommended next action
Reported SLA compliance is 17.64%.	Review SLA targets by priority, confirm the definition of an eligible ticket, and segment breaches by priority, category, region, and equipment.
5,318 tickets took more than 12 hours to resolve.	Review long-running tickets and group them by incident category, equipment type, priority, and ownership to identify recurring blockers.
2,753 tickets are open or in progress.	Review backlog age, priority, owner, and time since last update.
Connectivity and Intermittency contribute high volume.	Investigate repeat incidents and common patterns before proposing technical fixes or preventive actions.
August 2026 is a partial month.	Avoid comparing August directly with full months; use comparable date ranges or mark the month as partial.

Limitations and validation notes

• The dataset ends on August 14, 2026, so August is incomplete.

• Null resolution times are not zero hours and cannot be evaluated using MTTR alone.

• The reported SLA compliance rate applies to tickets classified as compliant or breached in the current model, not all 10,020 tickets.

• The equipment visual includes a blank category; some tickets do not match an equipment record, so equipment-based conclusions should account for unmatched IDs.

• The analysis is descriptive. Root-cause and staffing recommendations require follow-up analysis and operational context.

7. Conclusion

NOC Performance Analytics consolidates ticket demand, status, resolution time, equipment and category distribution, and SLA outcomes. The project demonstrates how Databricks SQL and Power BI can turn operational records into an interactive monitoring and analysis tool.

The strongest signals are the low reported SLA compliance rate, the large number of tickets taking more than 12 hours to resolve, and the 2,753 tickets that remain open or in progress. Connectivity and Intermittency account for a substantial share of workload and should be investigated for recurring patterns.

The recommended next step is to validate SLA definitions, analyze the drivers of breaches, review aging backlog, and investigate long-resolution tickets by priority and incident category. This moves the report from descriptive monitoring toward actionable operational improvement.

Project summary

• **Business objective:** understand ticket volume, resolution performance, and SLA compliance.

• **Data source:** Databricks SQL; tickets and equipos in the Power BI model.

• **Tools:** Databricks SQL, Power Query, Power BI data modeling, DAX, and Power BI visuals.

• **Deliverable:** a three-page interactive dashboard and this case-study document.

• **Business value:** better visibility into workload, resolution patterns, and areas requiring further investigation.

NOC PERFORMANCE ANALYTICS

Databricks SQL • Power Query • Power BI • DAX